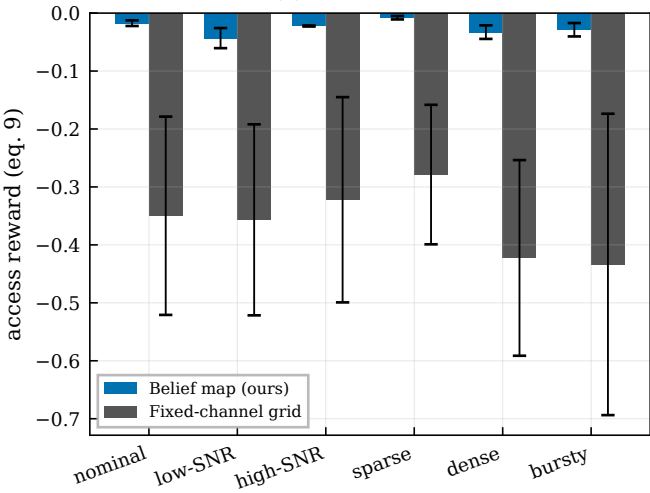


(a) access reward



(b) PU collision

